

Econ 30065: Microeconomics

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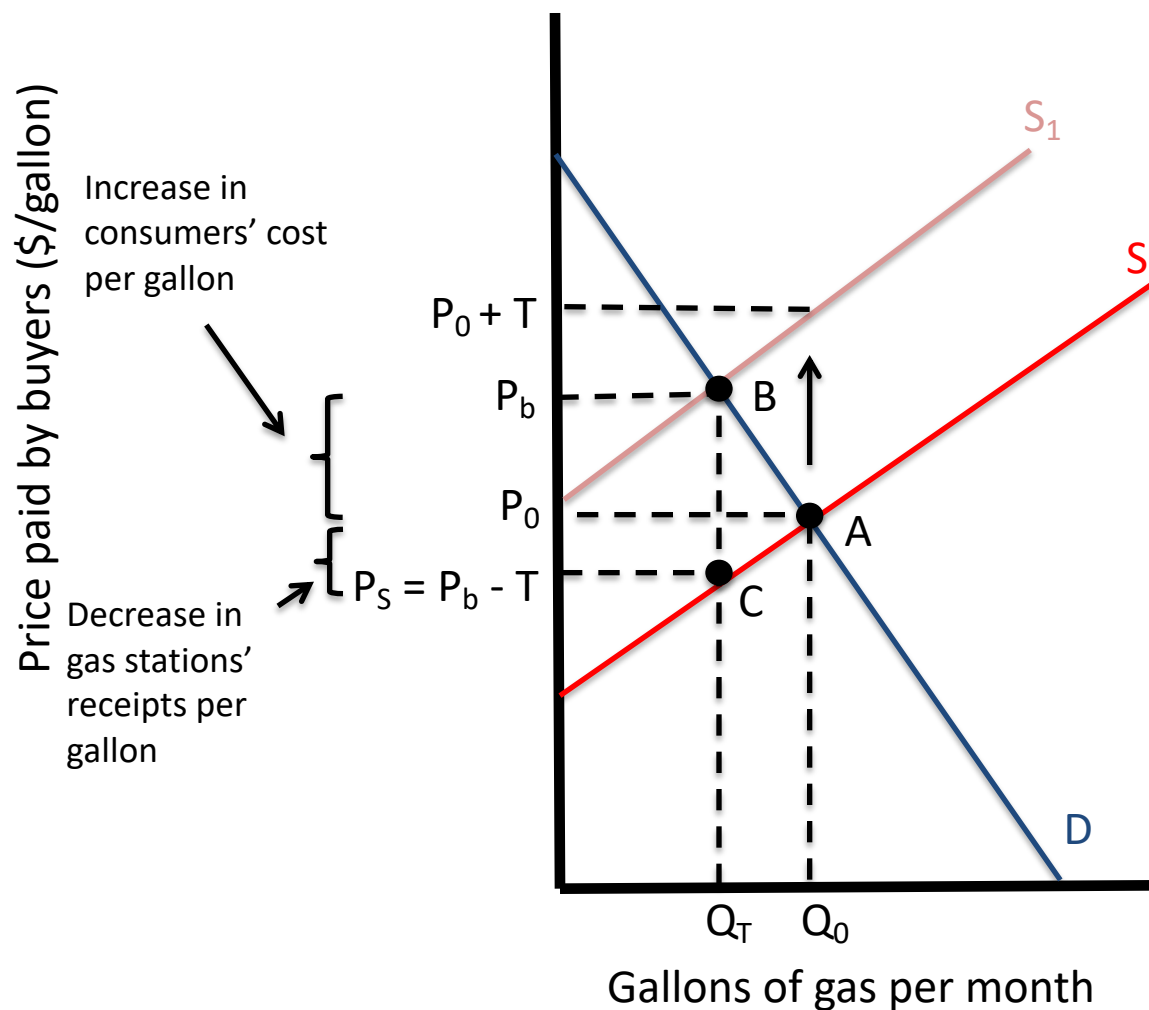
Market intervention

Taxes

- Governments tax goods to raise the revenue needed to pay public expenditures
- Taxation alters the competitive equilibrium
- There are different kinds of taxes. We will focus on:
 - **Specific tax:** a fixed dollar amount that must be paid on each unit bought or sold
 - Example: 3\$ per unit

Effects of a specific tax T on sellers

- Supply curve shifts left by T
- Point A shows the market equilibrium
- Point B shows the increase in the price paid by consumers due to the tax
- Point C shows the decrease in the price received by producers due to the tax
- Area $P_b - P_s - B - C$: total taxes collected.



Effects of a specific tax T on sellers

- For any price paid by consumers, firms now receive less than when there is no tax
 - Won't be willing to supply as much as before
 - Supply curve with the tax is a distance T above the original supply curve
- New equilibrium price paid by consumers is price at which the demand curve and new supply curve cross
 - Amount bought and sold falls
 - Price paid by consumers rises; price received by firms falls

Example

- The demand for hotel rooms in Mallorca is given by $Q_D = 10 - p$ (where p denotes the price of a hotel room). The supply of hotel rooms is given by $Q_S = p$.
- Determine the equilibrium price and the equilibrium quantity for this market both graphically and analytically. Label your graph carefully.
- $Q_S = Q_D \rightarrow p = 10 - p \rightarrow p^* = 5 \rightarrow Q^* = 5.$

Example

- Mallorca's prime minister plans to introduce a tax $t = 2$ on each hotel room on producers. Determine the new (after tax) equilibrium quantity. How much will consumers have to pay for a hotel room after the introduction of the tax? What will be the price received by producers for each unit supplied?
 - $P_s + t = P_b$
 - $Q + 2 = 10 - Q$
 - $Q^* = 4.$
 - $P_b = 6$ (*price paid by consumers, so consumers now pay 1 more per unit sold*).
 - $P_s = 4$ (*price received by producers, so producers now receive 1 less per unit sold*).

Incidence of a specific Tax

- **Incidence of a tax:** how much of the tax burden is borne by various market participants.
- From the graph we see that the burden of the tax is shared by consumers and producers

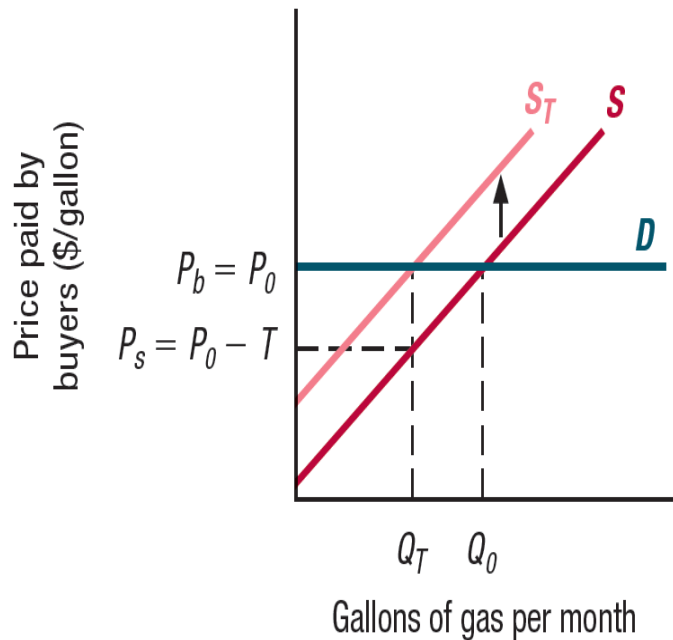
Incidence of a specific Tax

- **Incidence of a tax:** how much of the tax burden is borne by various market participants.
 - It depends on elasticity of demand and supply curves
- 4 cases:
 - Perfectly elastic demand: sellers bear the entire burden of the tax
 - Perfectly inelastic supply: sellers bear the entire burden of the tax
 - Perfectly inelastic demand: buyers bear the entire burden of the tax
 - Perfectly elastic supply: buyers bear the entire burden of the tax

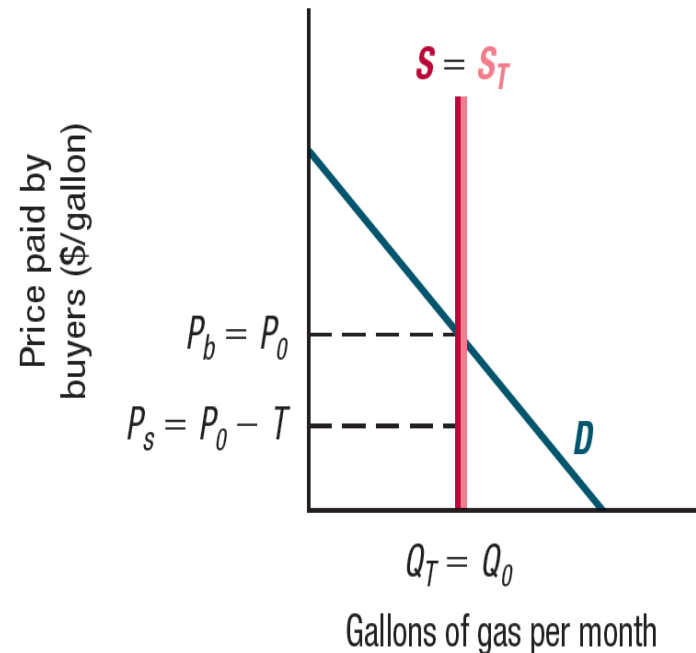
Incidence of a Specific Tax

- Sellers bear the entire burden of the tax

(a) Perfectly elastic demand



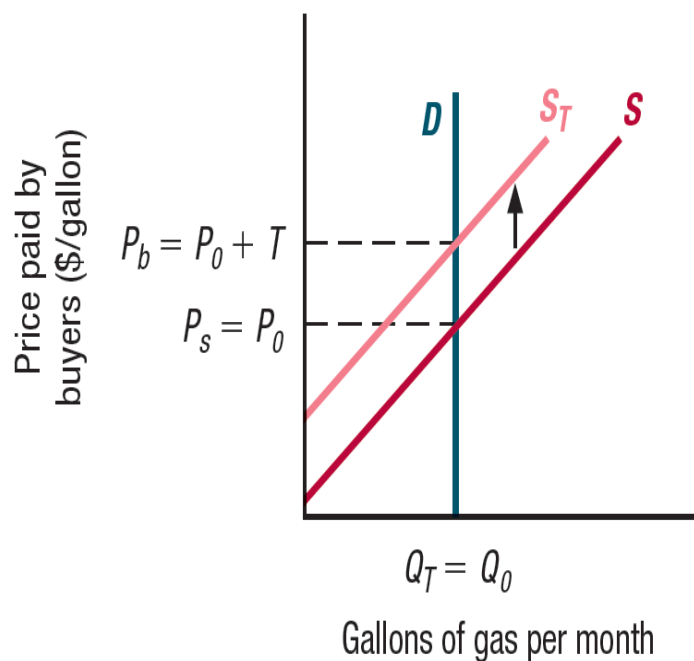
(b) Perfectly inelastic supply



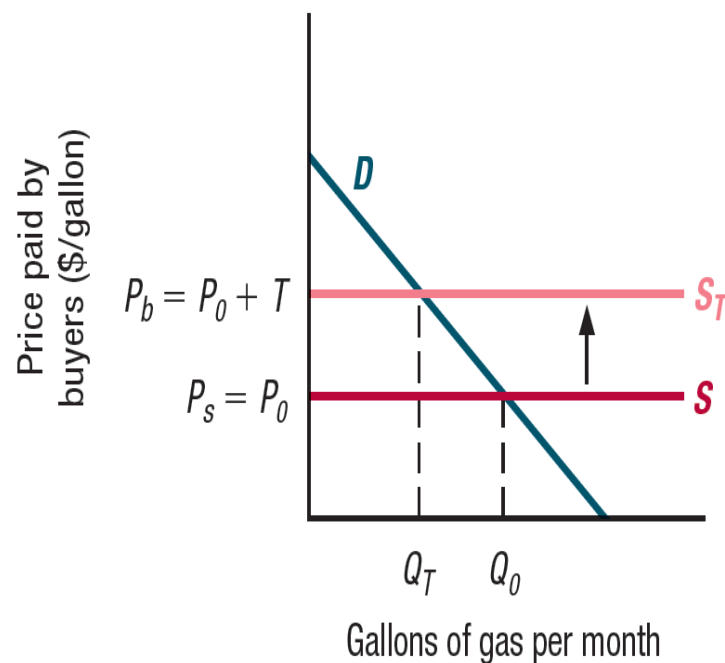
Incidence of a Specific Tax

- Buyers bear the entire burden of the tax

(c) Perfectly inelastic demand



(d) Perfectly elastic supply



Incidence of a specific Tax

- The more elastic demand is, and the less elastic supply is, more of the tax is borne by seller
- For small taxes:

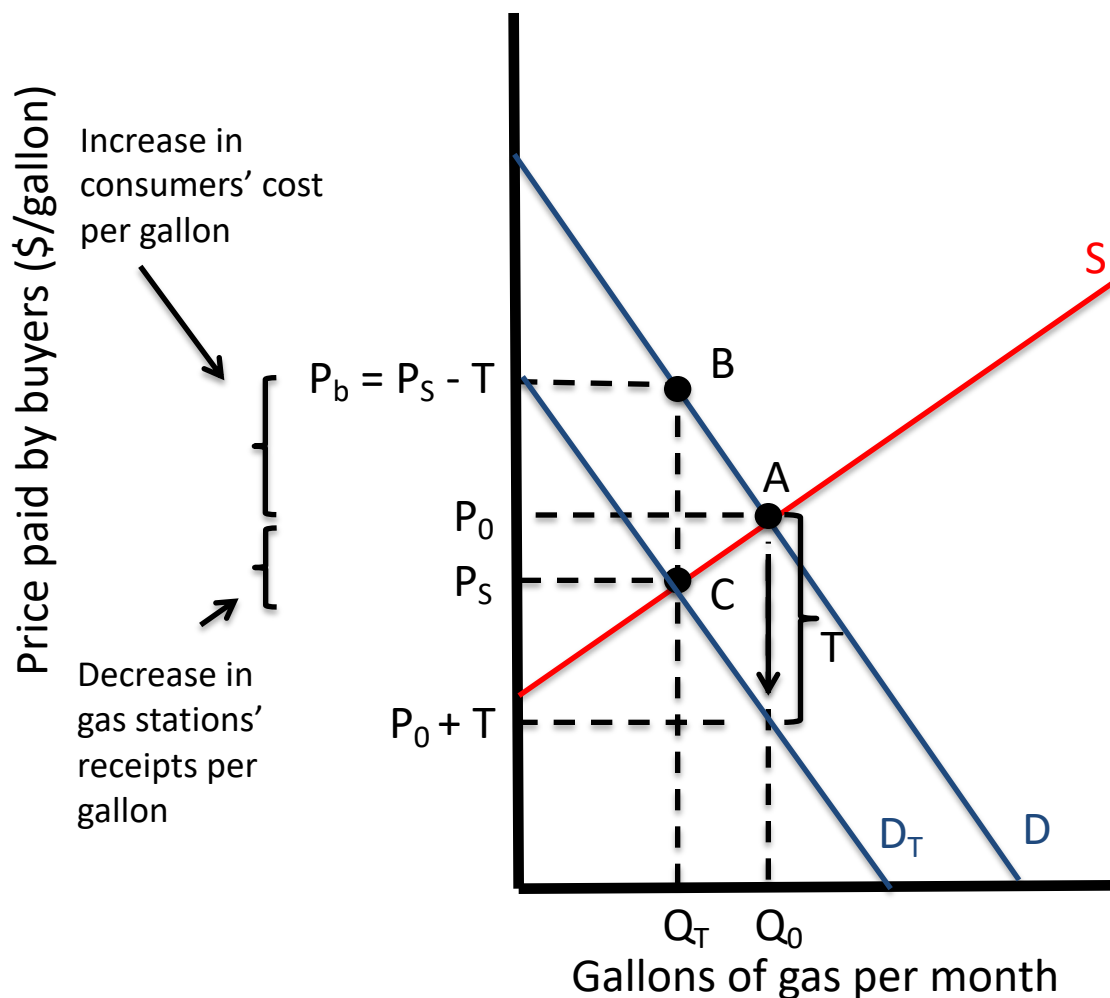
$$\text{Buyers' share} = \frac{E^s}{E^s - E^d}$$

=>When $E^S = -E^D$ they share equally the burden

=>When supply is more elastic than demand: more burden on buyers

Effects of a specific tax T on buyers

- Demand shifts left by T
- Point A shows the market equilibrium
- Point B shows the increase in the consumer's cost due to the tax: $P_s + T$
- Point C shows the decrease in the producer's revenue per gallon due to the tax

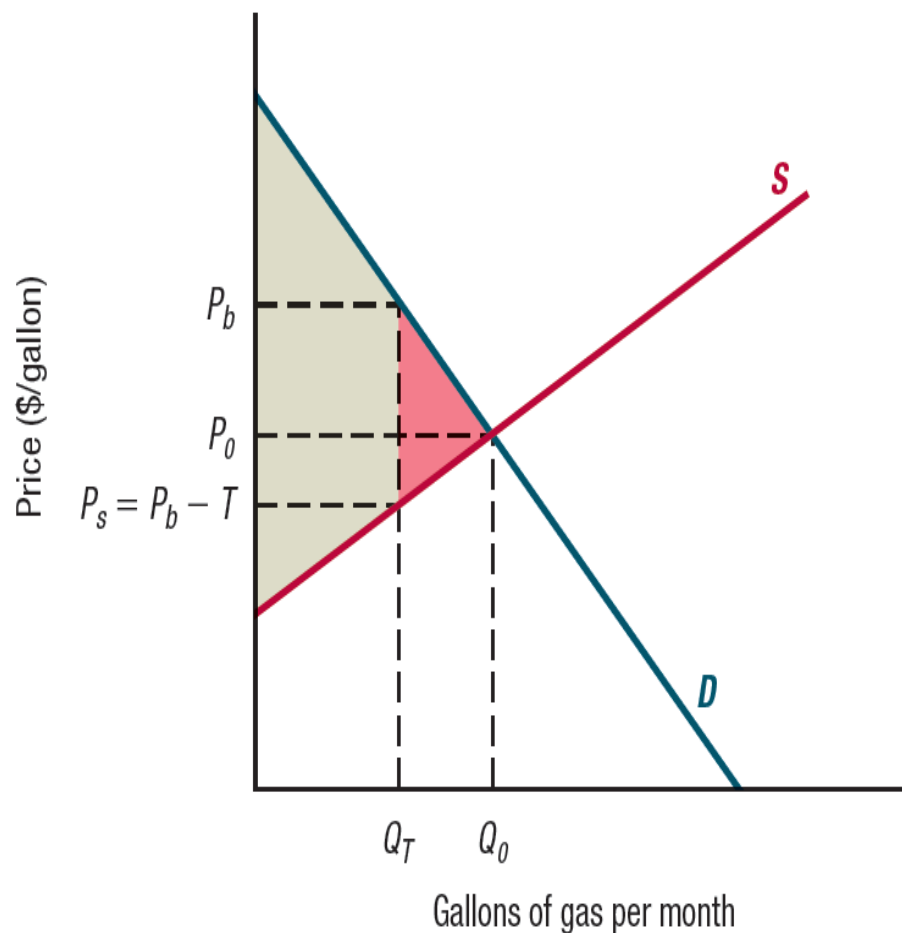


Effects of a specific tax T

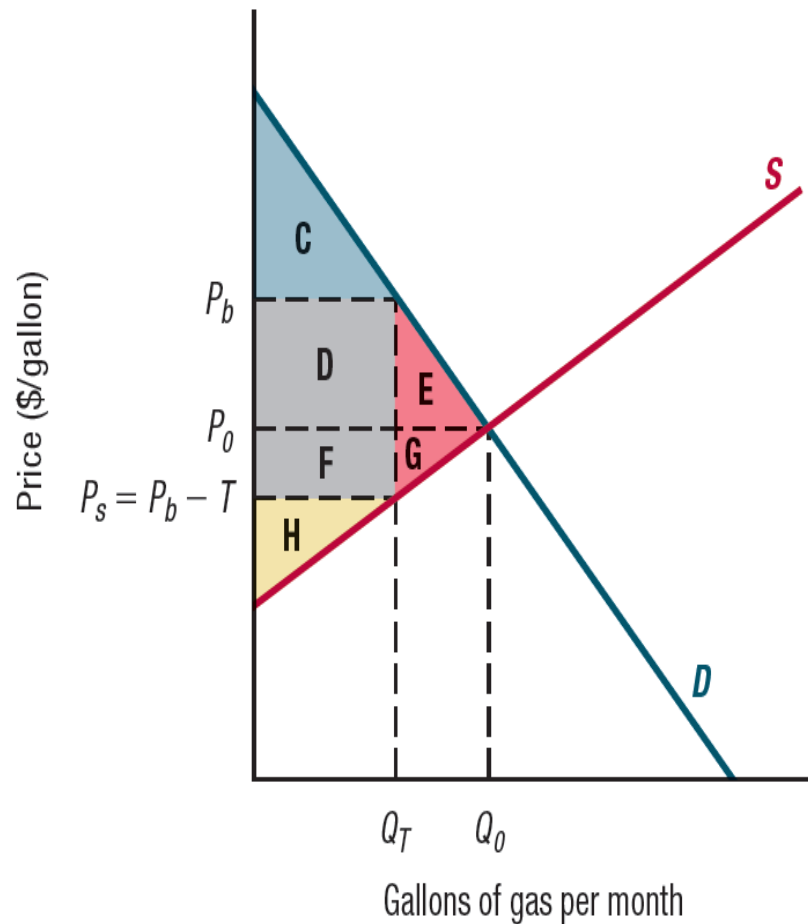
- In perfectly competitive markets, regardless of who is taxed by the government, the incidence remains the same
- Makes no difference whether the tax is levied on consumers or producers

The Welfare effects of a specific tax

Deadweight loss of taxation: lost aggregate surplus due to a tax



Welfare Effects of a Specific Tax



	Without tax	With tax	Gain/Loss with tax
Aggregate surplus	$C + D + E + F + G + H$	$C + D + F + H$	$E + G$
Consumer surplus	$C + D + E$	C	$D + E$
Producer surplus	$F + G + H$	H	$F + G$
Tax revenue	0	$D + F$	$D + F$